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Github files link:

Step 1: https://github.com/mrunalikatta1998/Data-Warehouse-and-Pipelines/blob/main/HW%206/etl_s3.py

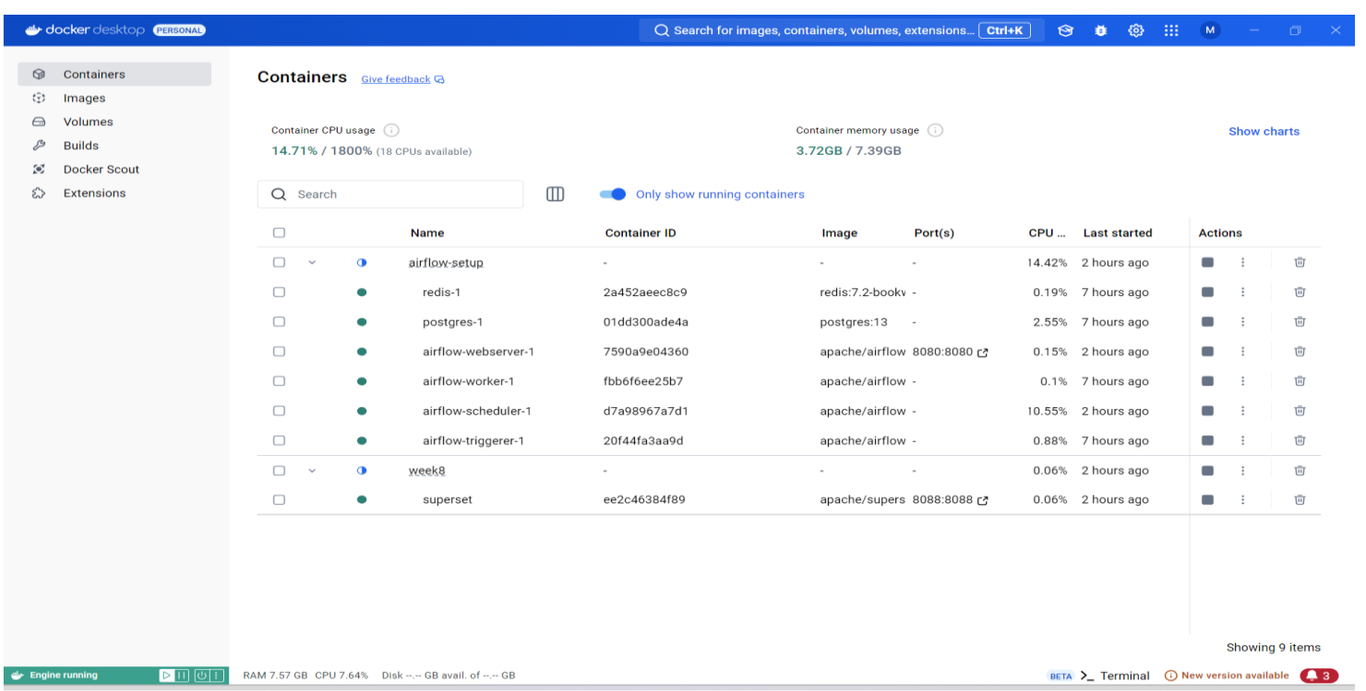
Step 2: https://github.com/mrunalikatta1998/Data-Warehouse-and-Pipelines/blob/main/HW%206/BuildELT_CTAS.py

Homework 06

1. Import two tables in your SnowflakeLinks to an external site. as an ETL DAG in your Airflow (+3pt)
    1. user_session_channel and session_timestamp (under raw schema or equivalent)
    2. Capture the screenshot of this DAG's detailed page from the Web UI (#1)

→ Docker

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mruna>git clone https://github.com/keeyong/sjsu-data226-SP25.git
Cloning into 'sjsu-data226-SP25'...
remote: Enumerating objects: 343, done.
remote: Counting objects: 100% (127/127), done.
remote: Compressing objects: 100% (122/122), done.
remote: Total 343 (delta 59), reused 0 (delta 0), pack-reused 216 (from 1)
Receiving objects: 100% (343/343), 129.65 KiB | 1.44 MiB/s, done.
Resolving deltas: 100% (184/184), done.

C:\Users\mruna>cd sjsu-data226-SP25\week8

C:\Users\mruna\sjsu-data226-SP25\week8>docker compose up
[+] Running 3/3
 ✔Network week8_superset_network   Created                0.1s
 ✔Container week8-superset_db-1    Created                0.2s
 ✔Container superset               Created                0.1s
Attaching to superset, superset_db-1
superset_db-1  | The files belonging to this database system will be owned by user "postgres".
superset_db-1  | This user must also own the server process.
superset_db-1  |
superset_db-1  | The database cluster will be initialized with locale "en_US.utf8".
superset_db-1  | The default database encoding has accordingly been set to "UTF8".
superset_db-1  | The default text search configuration will be set to "english".
superset_db-1  |
superset_db-1  | Data page checksums are disabled.
superset_db-1  |
superset_db-1  | initdb: error: directory "/var/lib/postgresql/data" exists but is not empty
superset_db-1  | It contains a dot-prefixed/invisible file, perhaps due to it being a mount point.
superset_db-1  | Using a mount point directly as the data directory is not recommended.
superset_db-1  | Create a subdirectory under the mount point.
superset_db-1 exited with code 1
superset       | Defaulting to user installation because normal site-packages is not writeable
superset       | Collecting snowflake-sqlalchemy (from -r /app/pythonpath/requirements-local.txt (line 1))
```

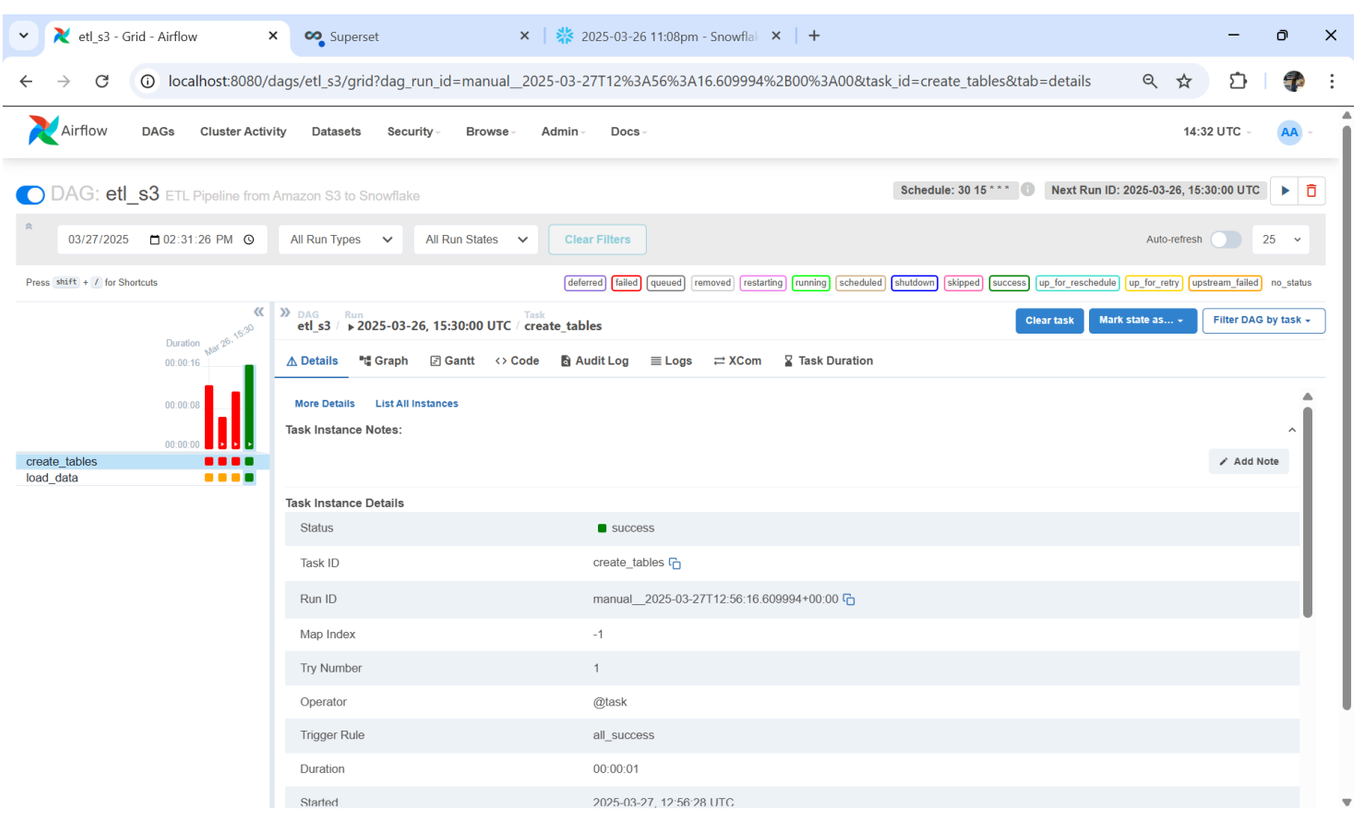


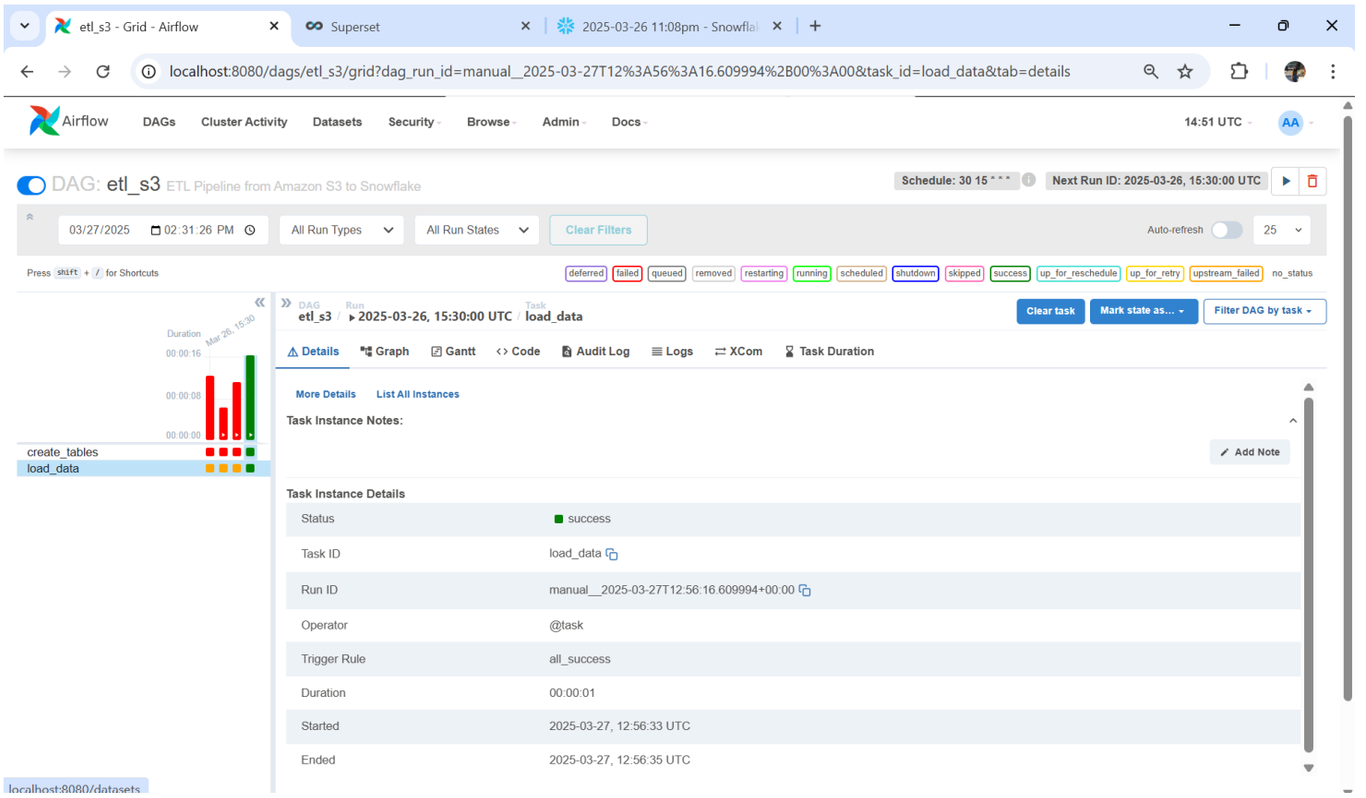
Below is the code for ETL pipeline to create 2 tables in my snowflake:

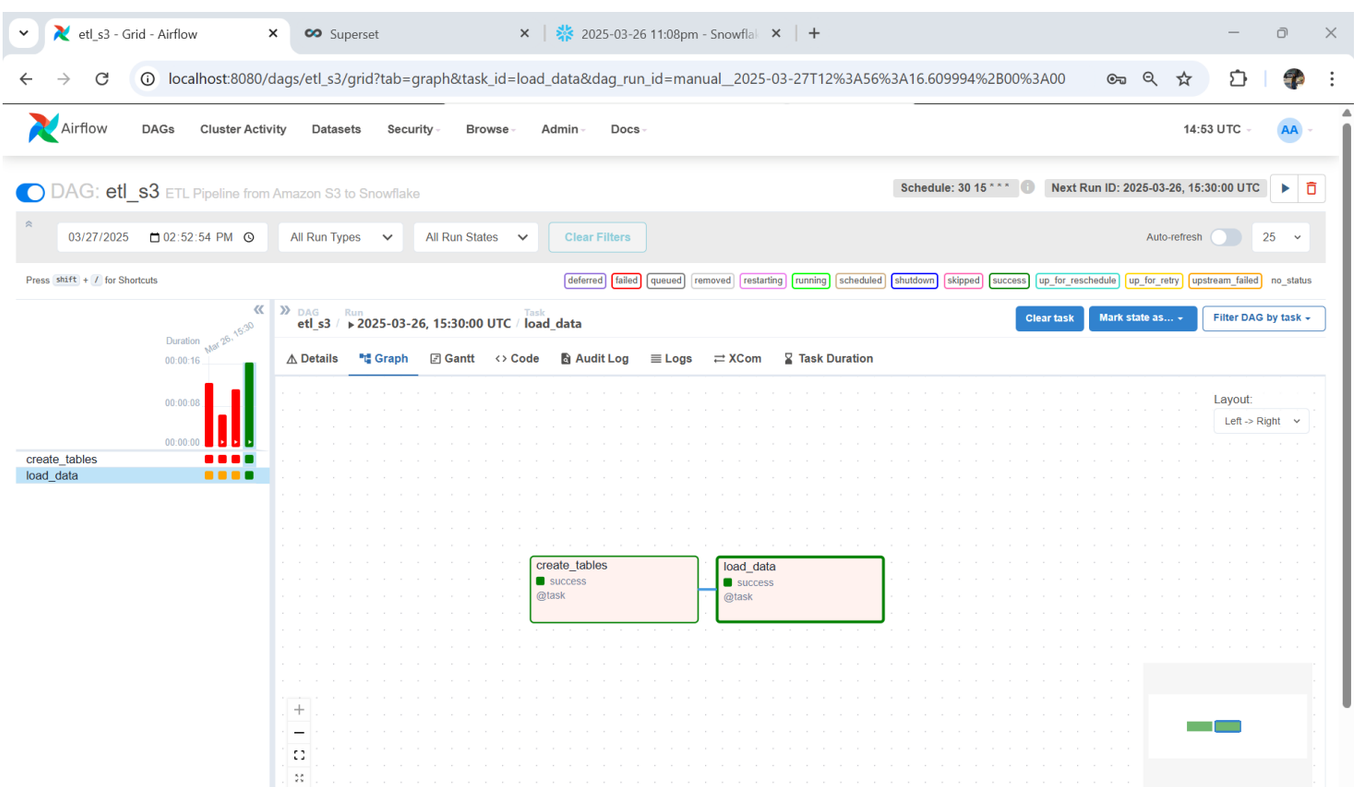
etl_s3.py

```
from airflow import DAG
from airflow.models import Variable
from airflow.decorators import task
from airflow.operators.python import get_current_context
from airflow.providers.snowflake.operators.snowflake import SnowflakeOperator
from airflow.providers.snowflake.hooks.snowflake import SnowflakeHook
from datetime import datetime
from datetime import timedelta
import logging
import snowflake.connector

# Snowflake connection function
def return_snowflake_conn():
    hook = SnowflakeHook(snowflake_conn_id='snowflake_conn')  # 'snowflake_conn' should be defined in Airflow
connections
    conn = hook.get_conn()
    return conn.cursor()

# Task to create tables
@task
def create_tables(con, table1, table2):
    try:
        con.execute("BEGIN")
        con.execute(f"""
            CREATE TABLE IF NOT EXISTS {table1} (
                userId INT NOT NULL,
                sessionId VARCHAR(32) PRIMARY KEY,
                channel VARCHAR(32) DEFAULT 'direct'
            );
        """)
        con.execute(f"""
            CREATE TABLE IF NOT EXISTS {table2} (
                sessionId VARCHAR(32) PRIMARY KEY,
                ts TIMESTAMP
            );
        """)
        con.execute("COMMIT")
        return table1, table2
    except Exception as e:
        con.execute("ROLLBACK")
        print(e)
        raise(e)

# Task to load data from S3 to Snowflake
@task
def load_data(tables, url):
    try:
        con.execute("BEGIN")
        con.execute(f"""
            CREATE OR REPLACE STAGE dev.raw.blob_stage
            URL = '{url}'
            FILE_FORMAT = (TYPE = CSV, SKIP_HEADER = 1, FIELD_OPTIONALLY_ENCLOSED_BY = '');
        """)
        con.execute(f"""
            COPY INTO {tables[0]} FROM @dev.raw.blob_stage/user_session_channel.csv;
        """)
        con.execute(f"""
```

```
        COPY INTO {tables[1]} FROM @dev.raw.blob_stage/session_timestamp.csv;
    """)
    con.execute("COMMIT")
  except Exception as e:
    con.execute("ROLLBACK")
    print(e)
    raise(e)

# Define the DAG
with DAG(
  dag_id='etl_s3',  # The ID for the DAG
  description='ETL Pipeline from Amazon S3 to Snowflake',  # Description of the DAG
  start_date=datetime(2024, 10, 6),  # Start date of the DAG
  catchup=False,  # Don't backfill the DAG
  tags=['ETL'],  # Tags for categorizing the DAG
  schedule_interval='30 15 * * *'  # Schedule the DAG to run at 3:30 PM UTC daily
) as dag:

  # Initialize Snowflake connection
  con = return_snowflake_conn()

  # Get the URL from Airflow's Variable
  url = Variable.get("url")  # Ensure the "url" variable is set in Airflow

  # Define table names
  table1 = "dev.raw.user_session_channel"
  table2 = "dev.raw.session_timestamp"

  # Create the tables
  tables = create_tables(con, table1, table2)

  # Load the data from S3 into Snowflake
  load = load_data(tables, url)
```







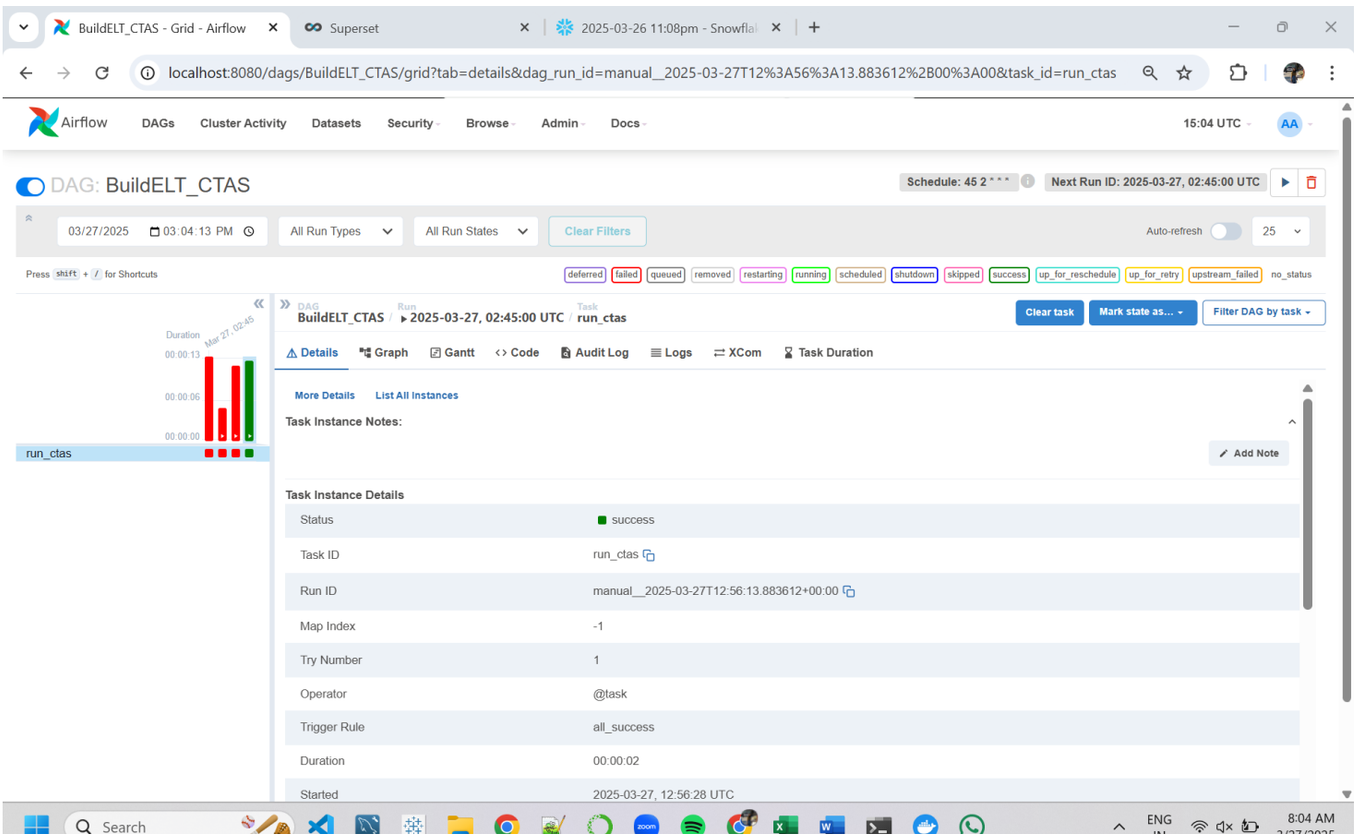
2. Create a ELT DAG in your Airflow to create a JOINED table of the two as described during the class (+3pt)
    1. session_summary (under analytics)
    2. **Extra point**: add one more condition to check duplicate records (+1pt)
    3. Capture the screenshot of this DAG's detailed page from the Web UI (#2)

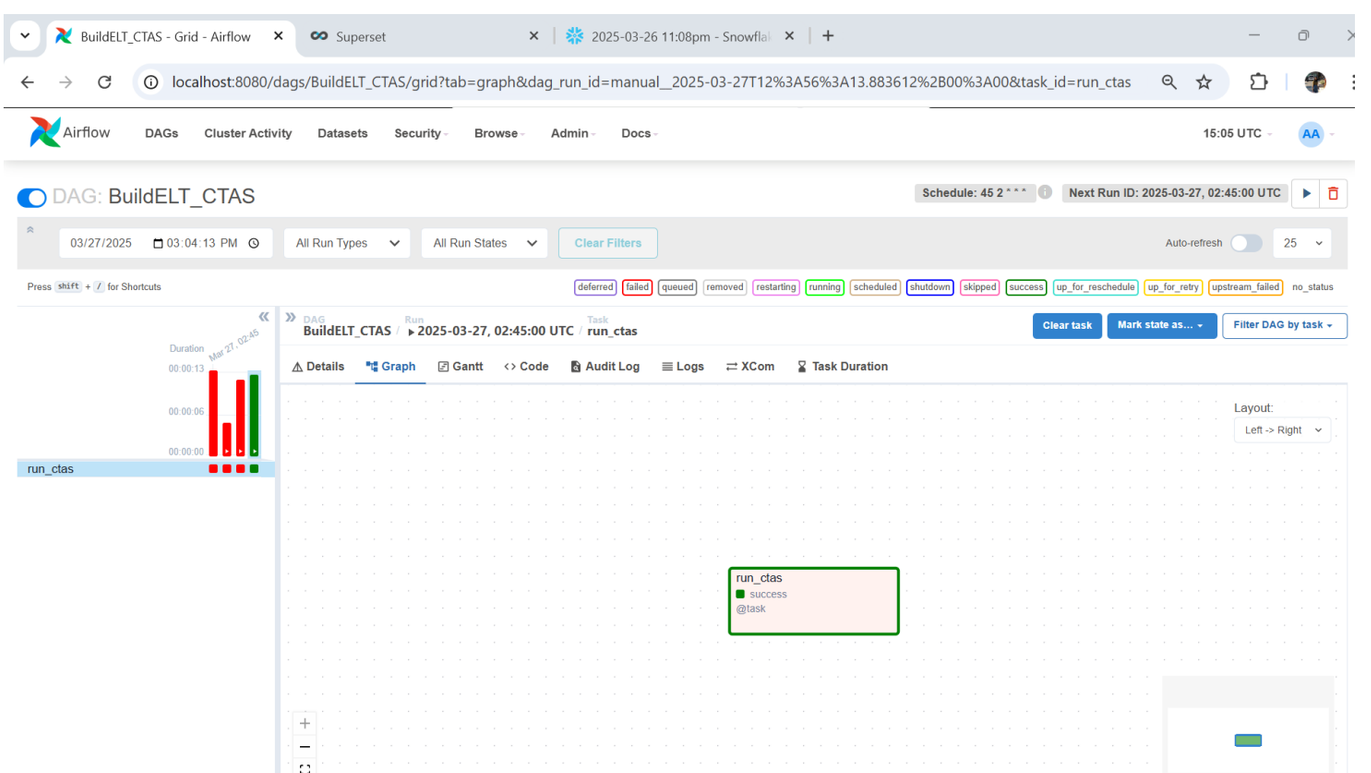
➔ Below is the code to create join of tables in analytics schema and condition to check for duplicate records:

BuildELT_CTAS.py

```
from airflow import DAG
from airflow.models import Variable
from airflow.decorators import task
from airflow.operators.python import get_current_context
from airflow.providers.snowflake.operators.snowflake import SnowflakeOperator
from airflow.providers.snowflake.hooks.snowflake import SnowflakeHook
from datetime import datetime
from datetime import timedelta
import logging
import snowflake.connector

# Function to return a Snowflake connection
def return_snowflake_conn():
    hook = SnowflakeHook(snowflake_conn_id='snowflake_conn')
    conn = hook.get_conn()
    return conn.cursor()

# Task to create or replace the table in Snowflake
@task
def run_ctas(table, select_sql, primary_key=None):
    logging.info(table)
    logging.info(select_sql)

    cur = return_snowflake_conn()

    try:
        cur.execute("BEGIN;")
        sql = f"CREATE OR REPLACE TABLE {table} AS {select_sql}"
        logging.info(sql)
        cur.execute(sql)

        # Check for duplicates based on the primary key
        if primary_key is not None:
            sql = f"SELECT {primary_key}, COUNT(1) AS cnt FROM {table} GROUP BY 1 ORDER BY 2 DESC LIMIT
1"
            logging.info(sql)
            cur.execute(sql)
            result = cur.fetchone()
            logging.info(f"Primary key check: {result}")

            if int(result[1]) > 1:
                logging.error("Primary key uniqueness failed!")
                raise Exception(f"Primary key uniqueness failed: {result}")

        cur.execute("COMMIT;")
    except Exception as e:
        cur.execute("ROLLBACK")
        logging.error(f'Failed to execute SQL. Completed ROLLBACK! Error: {e}')
        raise

# DAG definition for the ELT process
with DAG(
    dag_id='BuildELT_CTAS',
    start_date=datetime(2024, 10, 2),
    catchup=False,
    tags=['ELT'],
    schedule_interval='45 2 * * *'  # Customize this schedule as needed
) as dag:
    table = "dev.analytics.session_summary"
```

```
select_sql = """SELECT u.*, s.ts FROM dev.raw.user_session_channel u
                JOIN dev.raw.session_timestamp s ON u.sessionId = s.sessionId"""

# Run the task to create the joined table
run_ctas(table, select_sql, primary_key='sessionId')
```





3. Set up your Preset account or [Docker Superset environmentLinks to an external site.](#) or Tableau account (+2pt)
    1. This includes setting up Snowflake connection and import session_summary from step 2
    2. Capture the screenshot of your Datasets (#3)

➔ Below are the commands to setup Docker Superset Environment:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mruna\sjsu-data226-SP25\week8>docker exec -it superset superset db upgrade
/usr/local/lib/python3.10/site-packages/flask_limiter/extension.py:333: UserWarning: Using the in-memory storage for tracking rate limits as no storage was expli
/flask-limiter.readthedocs.io#configuring-a-storage-backend for documentation about configuring the storage backend.
  warnings.warn(
2025-03-27 13:15:28,347:INFO:superset.utils.screenshots:No PIL installation found
2025-03-27 13:15:29,391:INFO:superset.utils.pdf:No PIL installation found
WARNI [alembic.env] SQLite Database support for metadata databases will         be removed in a future version of Superset.
INFO  [alembic.env] Starting the migration scripts.
INFO  [alembic.runtime.migration] Context impl SQLiteImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.env] Migration scripts completed. Duration: 00:00:00

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug superset
    Learn more at https://docs.docker.com/go/debug-cli/

C:\Users\mruna\sjsu-data226-SP25\week8>docker exec -it superset superset init
/usr/local/lib/python3.10/site-packages/flask_limiter/extension.py:333: UserWarning: Using the in-memory storage for tracking rate limits as no storage was expli
/flask-limiter.readthedocs.io#configuring-a-storage-backend for documentation about configuring the storage backend.
  warnings.warn(
2025-03-27 13:15:38,924:INFO:superset.utils.screenshots:No PIL installation found
2025-03-27 13:15:39,618:INFO:superset.utils.pdf:No PIL installation found
2025-03-27 13:15:43,224:INFO:superset.security.manager:Syncing role definition
2025-03-27 13:15:43,312:INFO:superset.security.manager:Syncing Admin perms
2025-03-27 13:15:43,319:INFO:superset.security.manager:Syncing Alpha perms
2025-03-27 13:15:43,329:INFO:superset.security.manager:Syncing Gamma perms
2025-03-27 13:15:43,338:INFO:superset.security.manager:Syncing sql_lab perms
2025-03-27 13:15:43,346:INFO:superset.security.manager:Fetching a set of all perms to lookup which ones are missing
2025-03-27 13:15:43,352:INFO:superset.security.manager:Creating missing datasource permissions.
2025-03-27 13:15:43,359:INFO:superset.security.manager:Creating missing database permissions.
2025-03-27 13:15:43,364:INFO:superset.security.manager:Cleaning faulty perms

What's next:
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    Learn more at https://docs.docker.com/go/debug-cli/

C:\Users\mruna\sjsu-data226-SP25\week8>docker exec -it superset superset fab create-admin
/usr/local/lib/python3.10/site-packages/flask_limiter/extension.py:333: UserWarning: Using the in-memory storage for tracking rate limits as no storage was expli
/flask-limiter.readthedocs.io#configuring-a-storage-backend for documentation about configuring the storage backend.
  warnings.warn(
2025-03-27 13:17:03,354:INFO:superset.utils.screenshots:No PIL installation found
2025-03-27 13:17:04,303:INFO:superset.utils.pdf:No PIL installation found
Username [admin]: admin
User first name [admin]: admin
User last name [user]: admin
Email [admin@fab.org]: admin@admin.com
Password:
Repeat for confirmation:
Recognized Database Authentications.
Admin User admin created.

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug superset
    Learn more at https://docs.docker.com/go/debug-cli/

C:\Users\mruna\sjsu-data226-SP25\week8>
```
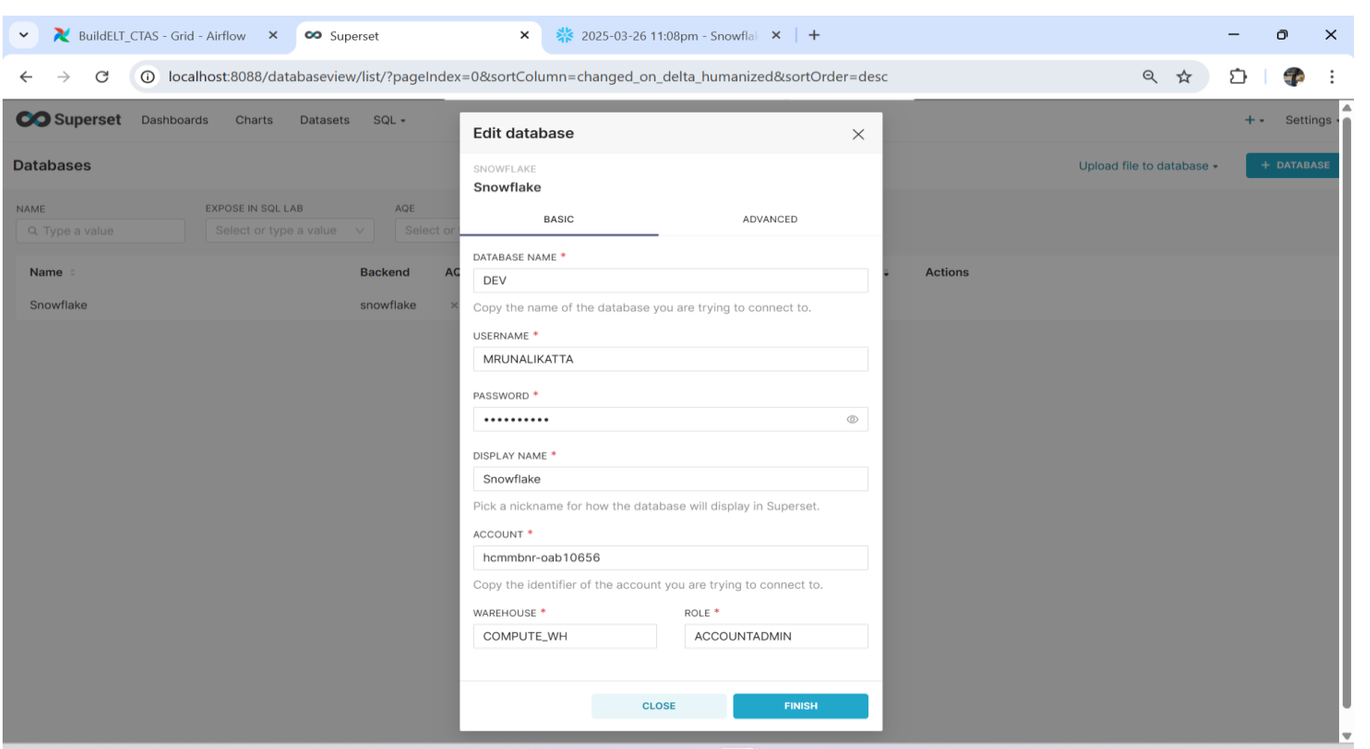
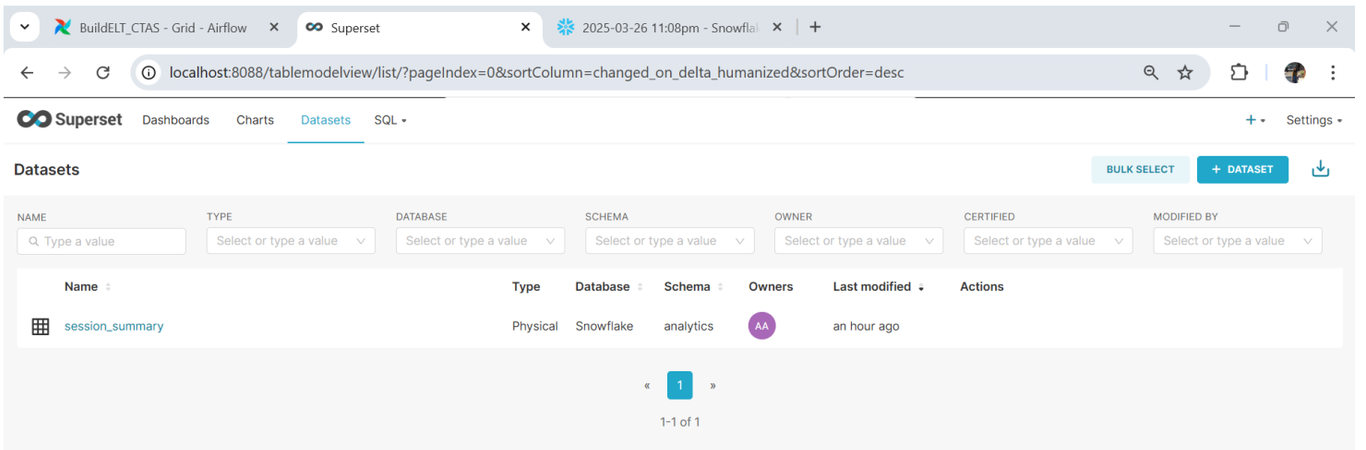

Below is the screenshot of SuperSet opened via Docker in which i am starting to connect my Snowflake account:

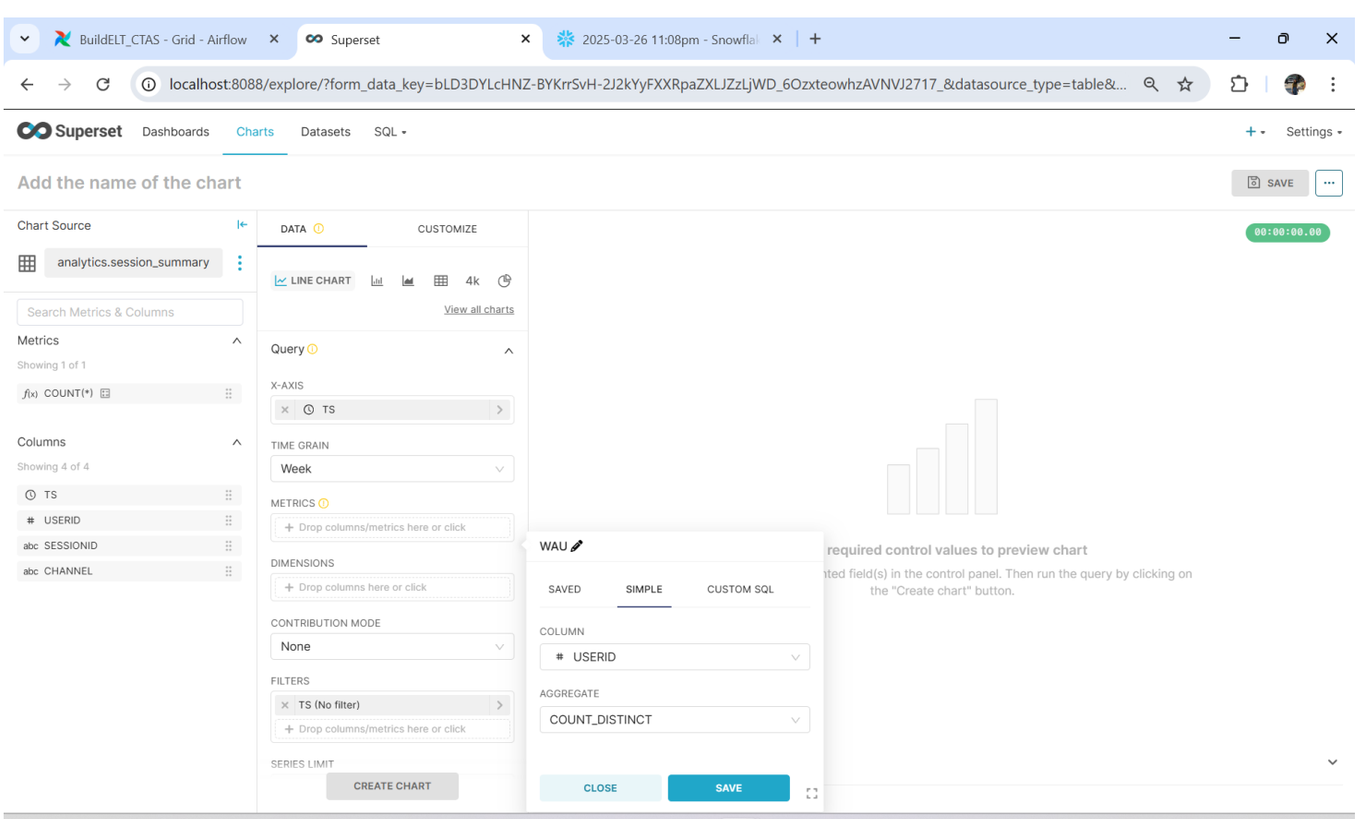


Below are the screenshots of my dataset:

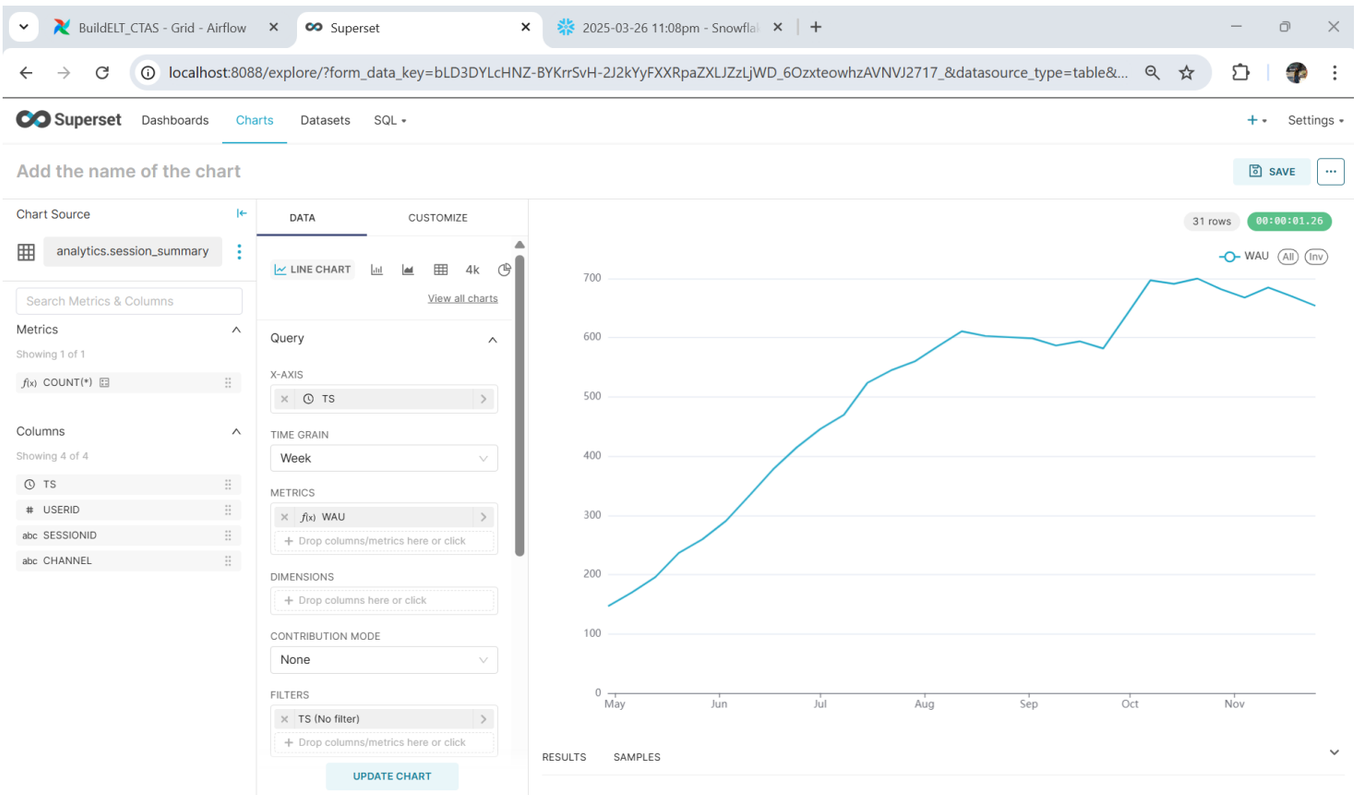


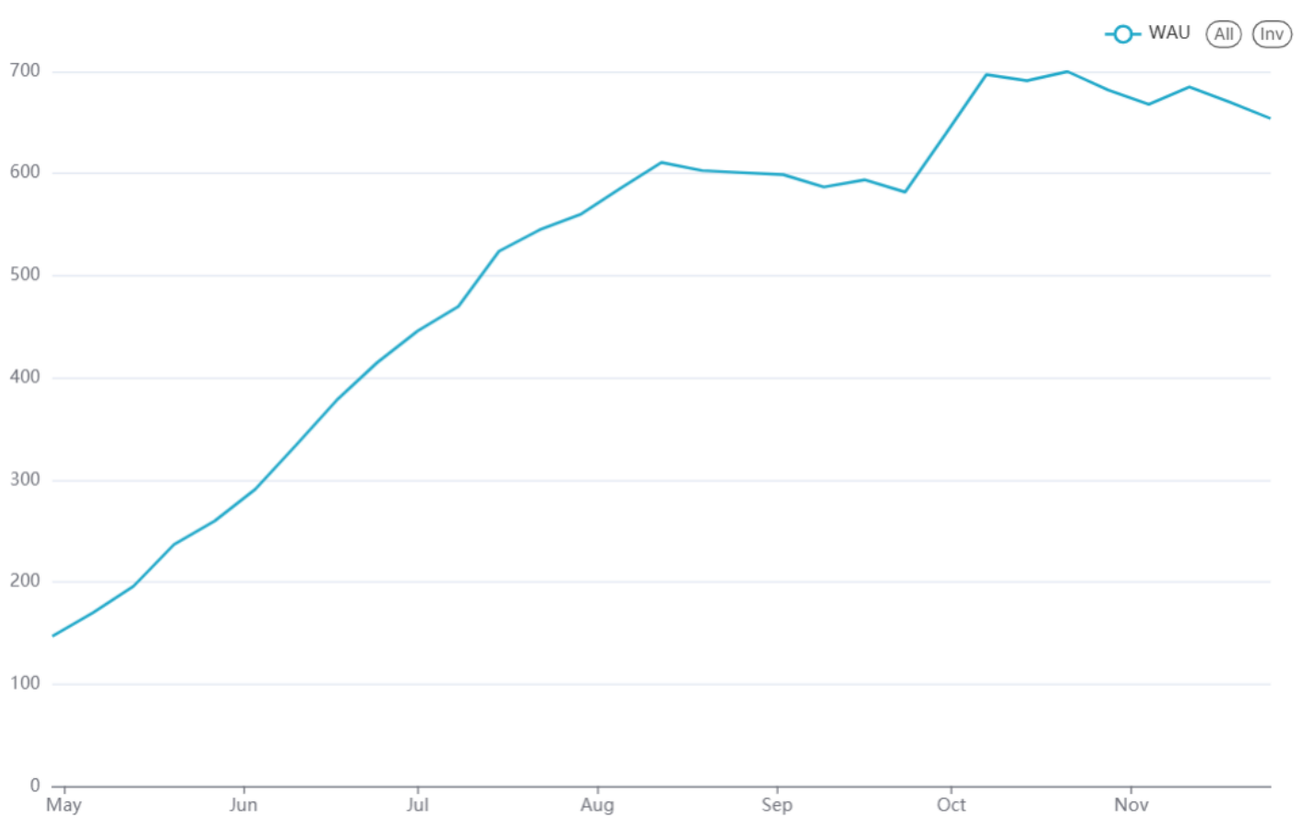
4. Create your WAU chart (+2pt)
    1. Make sure you rename the metrics field to **WAU**
    2. Capture the screenshot of the chart (#4)

➔ Below is the screenshot where I rename the metric field :



Line Chart


BuildELT_CTAS - Grid - Airflow
Superset
2025-03-26 11:08pm - Snowfla
localhost:8088/explore/?form_data_key=bLD3DYLcHNZ-BYKrrSvH-2J2kYyFXXRpaZXLJZzLjWD_6OzxteowhzAVNVJ2717_&datasource_type=table&...
Superset
Dashboards
Charts
Datasets
SQL
Settings
Add the name of the chart
SAVE
Chart Source
analytics.session_summary
Search Metrics & Columns
Metrics
Showing 1 of 1
f(x) COUNT(*)
Columns
Showing 4 of 4
TS
USERID
SESSIONID
CHANNEL
DATA
CUSTOMIZE
LINE CHART
4k
View all charts
Query
X-AXIS
TS
TIME GRAIN
Week
METRICS
f(x) WAU
Drop columns/metrics here or click
DIMENSIONS
Drop columns here or click
CONTRIBUTION MODE
None
FILTERS
TS (No filter)
Drop columns/metrics here or click
UPDATE CHART
31 rows
00:00:01.26
WAU
All
Inv
700
600
500
400
300
200
100
0
May
Jun
Jul
Aug
Sep
Oct
Nov
RESULTS
SAMPLES


WAU
All
Inv
700
600
500
400
300
200
100
0
May
Jun
Jul
Aug
Sep
Oct
Nov